

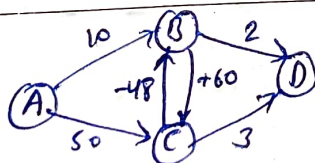
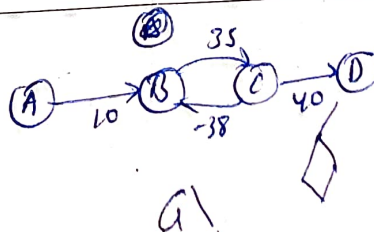
Department of Computer Science & Engineering, IIT (ISM), Dhanbad

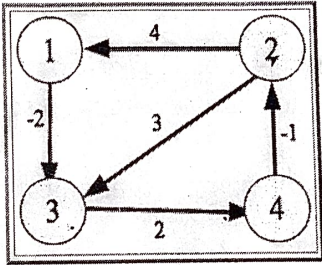
Department of Computer Science & Engineering, IIT (ISM), Dhanbad

Discipline: M.Tech. (CSE) & Research Scholar
Subject: Algorithmic Graph Theory (CSC503) **Time:** 2 hours, Marks: 50

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Instructions: Answer all questions

[illegible] $\frac{1}{2}$ 

	also mention the situation when Bellman-Ford Algorithm may fail (illustrate with an example).	
3	a. Illustrate with suitable example (considering a labelled tree T with at least 9 vertices) and converting T into its equivalent Prufer Sequence S and vice versa.	6
	b. Mention the important differences between BFS and DFS	4
4	a. Prove that Peterson Graph is non-planar using Kuratowski's and Wagner's Theorem as well.	6
	b. Prove that every planar graph has a dual.	4
5	a. Find the all pairs shortest paths among all vertices for the following graph using Floyd-Warshall Algorithm (showing all intermediate steps)	6
		
	b. Also describe Floyd-Warshall Algorithm to find the shortest paths among all vertices.	

Handwritten notes and calculations for the Floyd-Warshall Algorithm:

Initial Distance Matrix (D⁰):

	1	2	3	4
1	0	4	-2	
2		0	3	-1
3			0	2
4				0

Iteration 1 (k=1):

	1	2	3	4
1	0	4	-2	
2		0	3	-1
3			0	2
4				0

Iteration 2 (k=2):

	1	2	3	4
1	0	4	-2	
2		0	3	-1
3			0	2
4				0

Iteration 3 (k=3):

	1	2	3	4
1	0	4	-2	
2		0	3	-1
3			0	2
4				0

Iteration 4 (k=4):

	1	2	3	4
1	0	4	-2	
2		0	3	-1
3			0	2
4				0

Final Shortest Path Matrix (D⁴):

	1	2	3	4
1	0	4	-2	
2		0	3	-1
3			0	2
4				0

Handwritten notes:

- For (1,2): $(1,2) \rightarrow (1,3) \rightarrow (3,2)$ with weight -2 .
- For (1,4): $(1,4) \rightarrow (1,3) \rightarrow (3,4)$ with weight -2 .
- For (2,4): $(2,4) \rightarrow (2,3) \rightarrow (3,4)$ with weight -1 .
- For (3,4): $(3,4) \rightarrow (3,3) \rightarrow (3,4)$ with weight 2 .

End Semester Examination (Monsoon 2022-2023)
Department of Computer Science & Engineering, IIT (ISM), Dhanbad

Discipline: M.Tech. (CSE) & Research Scholar

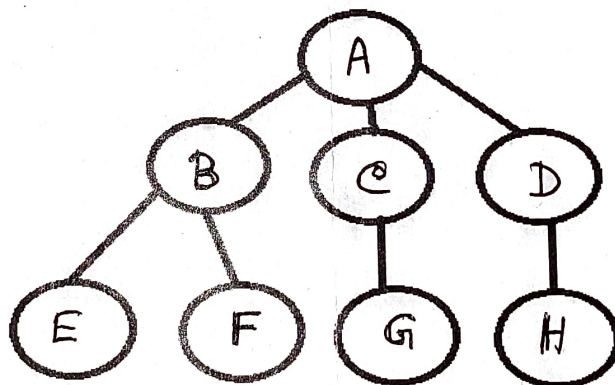
Subject: Algorithmic Graph Theory (CSC503)

Time: 3 hours, **Marks:** 100

Instruction: Answer all questions

Q. No		Marks
1.a	Why running time measurement is important to study? Mention some of the factors affecting the running time of a program.	(2+3)
b.	Deduce the Satisfiability problem in Clique decision problem or vice versa.	5
2.a	Prove that every graph G with maximum degree r is an induced subgraph of an r -regular graph.	4
b.	Write the O , Ω and θ notation for the function a) $n^2 \log n + n$ and b) $n!$.	(3+3)
3.a	Calculate the maximum flow for the following graph from source to sink node.	5
b.	Illustrate Kruskal's algorithm with a suitable example describing all steps to find the minimal spanning tree.	5
4.a	Given a graph and a source vertex in the graph, find the shortest paths from the source vertex (A) to all other vertices in the above graph using Dijkstra's shortest Path Algorithm.	5
b.	Illustrate the situation when Dijkstra's Algorithm may fail to find the shortest path but Bellman-Ford Algorithm may succeed. And also mention the situation when Bellman-Ford Algorithm may fail (illustrate with an example).	5

5.a	Find the all pairs shortest paths among all vertices for the graph (of Q. No. 4) using Floyd-Warshall Algorithm (showing all intermediate steps).	5
b.	Describe Floyd-Warshall Algorithm to find the shortest paths among all vertices.	5
6.a	Prove that Peterson Graph is non-planar using Kuratowski's Theorem.	4
b.	Prove that Peterson Graph is non-planar using Wagner's Theorem.	4
c.	Prove that every planar graph has a dual. ✓	2
7.a	For a planar graph G with n number of vertices and e number of edges with r regions, prove by method of induction that $n - e + r = 2$.	5
b.	The in-order and pre-order tree traversal of a given tree T is given by: In-order: D B H E I A F C J G K Pre-order: A B D E H I C F G J K Find the Prufer Sequence S of T (mentioning all steps).	5
8.a	If a graph G is maximal planar graph with n vertices ($n \geq 3$) and m edges then show that $m = 3n - 6$.	4
b.	Describe a method for finding a minimum connected dominating set and also, illustrate the same with suitable example.	3 + 3
9.a	The Prufer sequence of a tree T is given by: Prufer sequence: 1 7 6 6 1 Construct the tree T showing all steps for the same.	(2+3)
b.	Illustrate Prim's algorithm with an example.	5
10.a	Write algorithm for DFS & BFS.	5
b.	Trace DFS and BFS algorithm for the following graph.	5



End semester Examination (Monsoon 2023-24)
 Department of Computer Science & Engineering, IIT (ISM), Dhanbad
Discipline: M.Tech. (CSE) & Research Scholar
Subject: Algorithmic Graph Theory (CSC503) **Time: 3 hours, Marks: 100**

Instructions: Answer all questions

Monsoon Mid-Semester Examination, Session 2022-23

Examination & Semester: M.Tech. (Computer Science and Engineering) I Semester
Subject: Advanced DBMS (CSC502)

Instructions:

Time: 2 Hours
Max. Marks: 56

- (a) All questions are compulsory.
 (b) Attempt the questions in serial order.

Q. No.	Question	Marks
1. (a)	Consider the relational database <i>employee</i> (<u>person_name</u> , street, city) <i>works</i> (<u>person_name</u> , company_name, salary) <i>company</i> (<u>company_name</u> , city) <i>manages</i> (<u>person_name</u> , manager_name) Give an expression in the Relational Algebra (using Basic Operators only) for each request: (i) Modify the database so that Amit now lives in Dhanbad city. (ii) Give all employees of Facebook a 10 percent salary raise. (iii) Find the names of all employees in this database who live in the same city as the company for which they work.	6
(b)	Consider a relation $r(A, B, C, D, E, F, G)$ satisfies the following functional dependencies: $F = \{A \rightarrow B, BC \rightarrow DE, AEF \rightarrow G\}$. Find the closure of $\{AC\}$ under this set. Is functional dependency $ACF \rightarrow DG$ implied by F ? Show the complete steps.	4
2. (a)	Suppose you are given a relation R with four attributes $ABCD$. For each of the following sets of FDs, assuming those are the only dependencies that hold for R , do the following: (a) Identify the candidate key(s) for R . (b) Identify the best normal form that R satisfies (1NF, 2NF, 3NF, or BCNF). (c) If R is not in BCNF, decompose it into a set of BCNF relations that preserve the dependencies. Show each decomposition step (if any) clearly. (i) $C \rightarrow D, C \rightarrow A, B \rightarrow C$ (ii) $B \rightarrow C, D \rightarrow A$ (iii) $ABC \rightarrow D, D \rightarrow A$	6
(b)	A relation $r(A, B, C, D, E, F)$ has the following set of functional dependencies: $F = \{A \rightarrow CD, B \rightarrow C, F \rightarrow DE, F \rightarrow A\}$. The relation is decomposed into $r_1(A, B, C)$, $r_2(A, F, D)$, and $r_3(E, F)$. Check whether this decomposition is lossless or lossy?	4
3. (a)	Construct a B+-tree for the following set of key values: (2, 3, 5, 7, 11, 17, 19, 23, 29, 31) Assume that the tree is initially empty and values are added in ascending order. (I) Construct B+-trees for the cases where the number of pointers that will fit in one node is 6. (II) Show the steps involved in the following queries: (i) Find records with a search-key value of 11. (ii) Find records with a search-key value between 7 and 17, inclusive (III) Show the form of the tree after each of the following series of operations: (i) Insert 9 (ii) Insert 10 (iii) Insert 8 (iv) Delete 23 (v) Delete 19	8
(b)	Assume that only one tuple fits in a block and memory holds at most 3 page frames. Show the runs created on each pass of the sort-merge algorithm, when applied to sort the following tuples on the first attribute: (kangaroo, 17), (wallaby, 21), (emu, 1), (wombat, 13), (platypus, 3), (lion, 8), (warthog, 4), (zebra, 11), (meerkat, 6), (hyena, 9), (hornbill, 2), (baboon, 12).	4
4. (a)	Let relations $r_1(A, B, C)$ and $r_2(C, D, E)$ have the following properties: r_1 has 20,000 tuples, r_2 has 45,000 tuples, 25 tuples of r_1 fit on one block, and 30 tuples of r_2 fit on one block. Estimate the number of block accesses required, using each of the following join strategies for $r_1 \bowtie r_2$: (i) Nested-loop join. (ii) Block nested-loop join (iii) Merge join (iv) Hash join	8
(b)	What factors could result in skew when a relation is partitioned on one of its attributes by: (i) Hash partitioning (ii) Range partitioning In each case, what can be done to reduce the skew?	4

5. (a)	Consider join processing using symmetric fragment and replicate with range partitioning. How can you optimize the evaluation if the join condition is of the form $ r.A - s.B \leq k$, where k is a small constant. Here, $ x $ denotes the absolute value of x . A join with such a join condition is called a band join.	6
(b)	Describe a good way to parallelize each of the following.	6
	(i) The difference operation	
	(ii) Aggregation by the count operation	
	(iii) Aggregation by the count distinct operation	
	(iv) Aggregation by the avg operation	
	(v) Left outer join, if the join condition involves only equality.	
	(vi) Left outer join, if the join condition involves comparisons other than equality.	

DBMS: Quiz: 1 GATE (PYQ)

Quiz: 2 Korth book content

+
Mongrab

mid sem + end sem: Korth
Exercise
questions

End semester Examination (Monsoon 2023-24)
 Department of Computer Science & Engineering, IIT (ISM), Dhanbad
 Discipline: M.Tech. (CSE) & Research Scholar
 Subject: Algorithmic Graph Theory (CSC503) Time: 3 hours, Marks: 100

Instructions: Answer all questions

Q. No		Marks
1	a. Prove that for every graph G, $X(G) \leq V(G) + 1$. (Here $X(G)$ is the chromatic number of G) b. Prove that G is bipartite iff the chromatic number of G is 2.	5 5
2	a. The Prufer sequence S of a labelled tree T is $S = \{1, 7, 6, 6, 1\}$. Draw the tree T describing all the intermediate steps pictorially during the process. b. Write the algorithm of the same.	5 5
3	a. Prove by method of induction that a graph G is a tree iff G is acyclic and the number of edges m in G is equal to $n-1$, where n is the number of vertices in G. b. Deduce that clique decision problem is NP-Hard using Satisfiability problem.	5 5
4	a. Prove that for a simple graph G with n number of vertices ($n \geq 4$), and E number of edges and genus g satisfies $g \geq \lceil \frac{1}{8}(E - 3n) + 1 \rceil$ b. "A necessary and sufficient condition for a graph G to be planar is that for every circuit C of G the auxiliary graph $G^+(C)$ is bipartite", prove it.	5 5

$n \geq n-1-n$
 $n-1 \leq n-n$

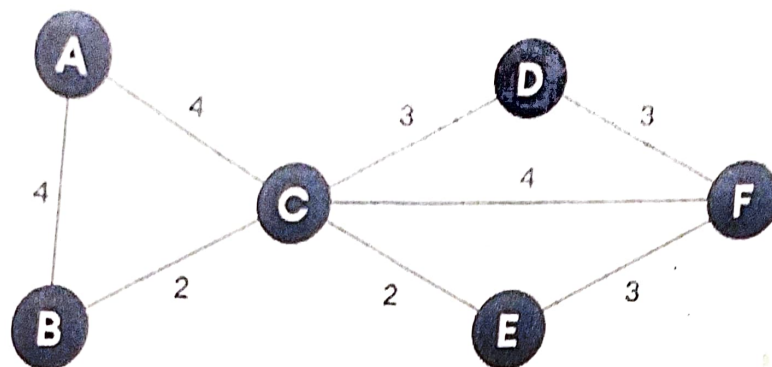
5	a. Suppose want to schedule some final exams for CS courses with following call numbers: CSE001, CSE002, CSE003, CSE004, CSE005, CSE006, CSE007, CSE008 and also suppose also that there are no common students in the following pairs of courses because of prerequisites: CSE001 - CSE002 CSE001 - CSE004 CSE001 - CSE003, CSE002 - CSE003 CSE001 - CSE007, CSE002 - CSE007 CSE001 - CSE008, CSE002 - CSE008, CSE003 - CSE008 CSE001 - CSE005, CSE002 - CSE005, CSE004 - CSE005, CSE001 - CSE006, CSE002 - CSE006, CSE004 - CSE006, CSE005 - CSE006 How many exam slots are necessary to schedule exams?	4
6	a. If a graph G is maximal planar graph with n vertices ($n \geq 3$) and m edges then show that $m = 3n - 6$.	5
	b. For a polyhedron with N number of vertices, E number of edges and F number of faces, prove by method of induction that $N - E + F = 2$.	5
7	a. Why running time measurement is important to study? Mention some of the factors affecting the running time of a program.	2+3
	b. Prove that every planar graph has a dual.	5
8	a. Discuss the merits and demerits of Dijkstra's Algorithm and Bellman-Ford Algorithm with examples.	5

$$\begin{aligned}
 m - n + 2 &= 2 \\
 m - n + 2 &= 2 \\
 n - m + 2 &= 2 \\
 2 &= 2
 \end{aligned}$$

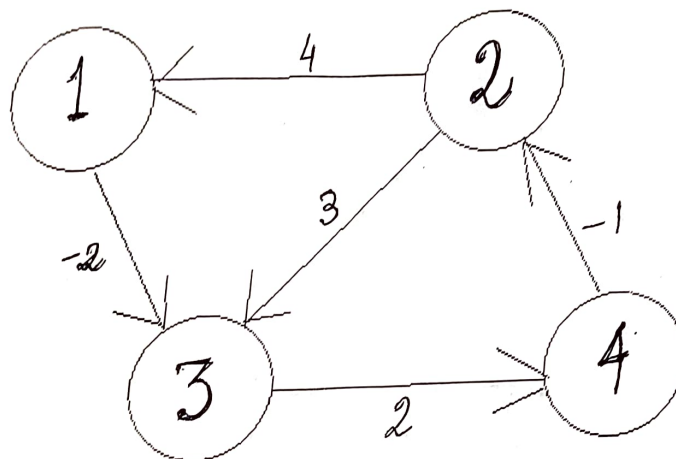
1, 2, 6, 4

5, 6

b. Use Prim's algorithm to find a minimum spanning tree that takes the following graph as input.



9 a. Find the all pairs shortest paths among all vertices for the following graph using Floyd-Warshall Algorithm (showing all intermediate steps)



a. Prove that Peterson Graph is non-planar using Kuratowski's and Wagner's Theorem as well.

Handwritten notes for the Floyd-Warshall algorithm:

Initial distance matrix (D⁽⁰⁾):

	1	2	3	4
1	0	4	-2	
2	4	0	3	-1
3	-2	3	0	2
4		-1	2	0

Intermediate steps (D⁽¹⁾, D⁽²⁾, D⁽³⁾) showing the progression of shortest paths as intermediate vertices are added.

Final shortest path matrix (D⁽⁴⁾):

	1	2	3	4
1	0	4	-2	1
2	4	0	3	-1
3	-2	3	0	2
4	1	-1	2	0

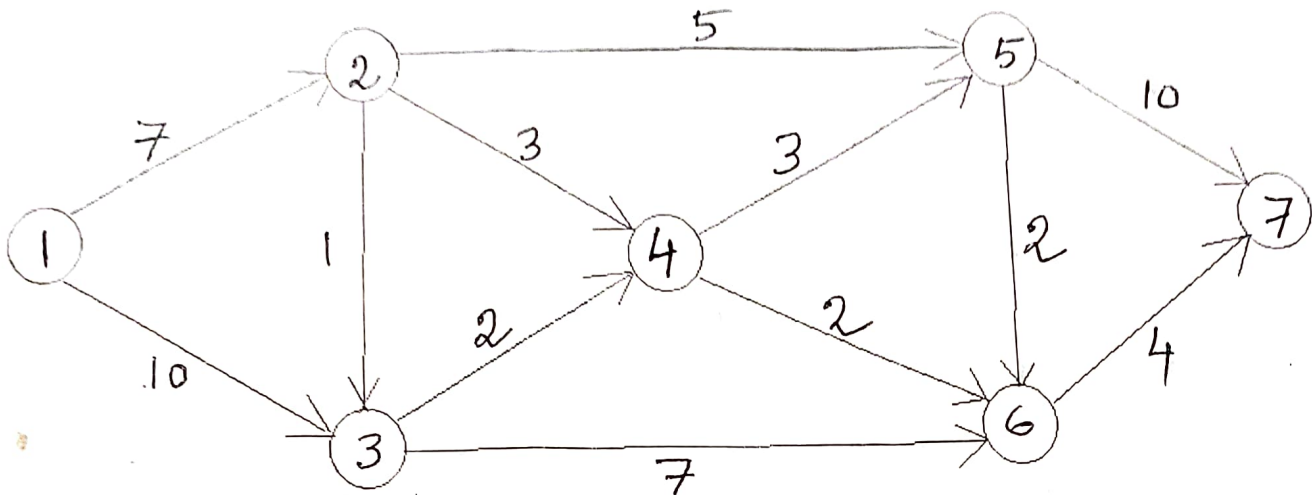
works (person_name, company_name, salary)

company (company_name, city)

manages (person_name, manager_name)

- (i) Modify the database so that Amit now lives in Dhanbad city.
- (ii) Give all employees of Facebook a 10 percent salary raise.

- 10 a. Find the maximum flow through the given network using Ford-Fulkerson algorithm.



- b. Consider the following graph to find the minimum connected dominating set showing all the intermediate steps during the constructions.

